

Amy Wu

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RESEARCH GOALS

I am interested in AI for accelerated scientific discovery. With skills in AI, mixed methods, and system design, my aim is to design autonomous AI systems for experimental design and scientific workflows in user facilities.

EDUCATION

University of Florida (UF)

Bachelor of Sciences with Honors in CS, Minor in Digital Arts and Sciences

Gainesville, FL

June 2020 – May 2025

PUBLICATIONS IN AI FOR ACCELERATED DISCOVERY

- **A. Wu**, J. De Chant, T. Hellert, S. Leemann, G. Martino, & A. Sulc. *Evaluating Language Models for Understanding Accelerators at Advanced Light Source*, Manuscript in Review. **International Particle Accelerator Conference (IPAC)'26**.
- T. Hellert, D. Abramov, M. Ajami, E. S. Barnard, W. Bhimji, G. Bhardwaj, T. Chavez, H. Daoud, T. J. Dunn, K. Fagnan, S. Farrell, J. G. F. M. Gazolla, L. Grbcic, D. Guenzing, J. Guo, Y. Ha, A. Hexemer, H. Hoschouer, X. Ju, K. Konate, G. Martino, D. McReynolds, M. O. Miller, S. A. Miskovich, P. Myint, S. Naseem, T. Northen, M. Patrou, G. E. Pérez, P. Quinn, H. Rafique, D. Ratner, A. Reed, E. Reyes, H. Shang, M. Smith, A. Sulc, C. Tennant, P. K. Tripathi, R. Underwood, M. Wall, D. Wang, **A. Wu**, C. Xu, K. G. Yager, & Z. Zhang. *Agentic AI for User Facilities*, in Synchrotron Radiation News. <https://doi.org/10.1080/08940886.2026.2649097>.
- T. Hellert, **A. Wu**, A. Sulc, C. Tennant, G. Martino, H. Shang, M. Smith, & Z. Zhang. *Status of Osprey: A Production Framework for Agentic AI in Control Systems*, Manuscript in Review. **International Particle Accelerator Conference (IPAC) '26**.

PUBLICATIONS IN MIXED METHODS (*ALL AUTHORS CONTRIBUTED EQUALLY.)

- **A. Wu**, M. Cobb, N. Graddy, V. Perez, C. Grant, & J. A. M. Waisome, *Guided Undergraduate Training for Shark Segmentation (GUTSS)*, in Special Interest Group on Computer Science Education (SIGCSE) 2024. <https://dl.acm.org/doi/10.1145/3626253.3635573>.
- A. DeHoog, J. Blanchard, **A. Wu**, & J. Hott. (2025, June), *Escaping the CS Dungeon: Modern College Curricula within and Beyond Computing*, Paper presented at 2025 American Society for Engineering Education (ASEE) Annual Conference & Exposition, Montreal, CN. <https://peer.asee.org/56440>.
- A. Webber, Z. Sayers, **A. Wu**, E. Thorner, J. Witter, G. Ayoubi, & C. Grant. 2024. *Analyzing Finetuned Vision Models for Mixtec Codex Interpretation*, in Workshop on Natural Language Processing for Indigenous Languages of the Americas (AmericasNLP) 2024. <https://aclanthology.org/2024.americasnlp-1.6.pdf>.
- C. Resch, P. Stapleton, B. Rheault, **A. Wu**, & C. Gardner-McCune. (2022, August), *Analysis of Effect of Answering Reflection Prompts in a Computer Organization Class*, Paper presented at 2022 American Society for Engineering Education (ASEE) Annual Conference & Exposition, Minneapolis, MN. <https://peer.asee.org/40428>.
- P. J. Sullivan, V. Chery*, E. R. Duarte*, R.F. Eloy*, D. Fabregat*, J. McCray*, J. K. Nelson*, Q. Nguyen*, S. B. Robles*, & **A. Wu.*** *Personal and Institutional Transformation in Higher Education through Mindfulness: Chapter 5.3. All Practice is Good Practice: Student Perspectives on Mental Health, Stigma and Mindfulness in Higher Education*, Manuscript in Publication. **Cambridge University Press**.

*All authors contributed equally.

RESEARCH EXPERIENCE (AI, MIXED METHODS (MM), SYSTEM DESIGN (SYS))

Sys Research Intern

PI: Dr. Drew Bertwistle, Mentor: Dr. Antonin Sulc @ ALS, Berkeley Lab

June 2026 - Present

Berkeley, CA

- Currently building a 3D magnetic model on a quadrupole magnet for beamline diagnostics.
- Assessing magnetic fields on BendA magnet as part of a digital twin.

Sys Research Assistant

PI: Dr. Drew Bertwistle, Mentor: Dr. Antonin Sulc @ ALS, Berkeley Lab

April 2026 - May 2026

Berkeley, CA

- Documented compiling Python and Mathematica versions for Radia.
- Built 3D magnetic model for a BendA magnet for the Advanced Light Source upgrade (ALS-U).
- **More info:** <https://als.lbl.gov/als-u/overview/>.

AI|Sys Department of Energy Intern

January 2026 - April 2026

PI: Dr. Thorsten Hellert, Mentor: Dr. Antonin Sulc @ ALS, Berkeley Lab

Berkeley, CA

- Assessed and visualized performance in commercial frontier, reasoning-specific, and LBL-hosted LLMs.
- Discussed, drafted, and edited a Department of Energy workshop report regarding agentic AI at user facilities.
- **Co-authorship:** <https://doi.org/10.1080/08940886.2026.2649097>.

AI|Sys Department of Energy Intern

May 2025 - August 2025

Dr. Yue Li (MSD) and Dr. Aikaterini Vriza (CNM), Argonne National Laboratory

Lemont, IL

- Selected specific materials from Materials Project database.
- Visualized the data through featurization techniques for UMAPs.
- Developed a machine learning model with featurization techniques to conduct property prediction.

AI|MM University Scholars Program (USP) AI Scholars Researcher

May 2023 – May 2025

Dr. Christan Grant and Dr. Jeremy Waisome, UF

Gainesville, FL

- Led a team of six in Guided Universal Training for Shark Segmentation (GUTSS), an Artificial Intelligence (AI) application to teach K-12 biology classrooms more on shark anatomy.
- Experimented with parameters in semantic segmentation algorithms (Meta's SAM and GroundingDINO).
- Established an application workflow incorporating semantic segmentation algorithms.
- Wrote instructor survey questions on AI and course curriculum for application design and usage.
- **pub:** [10.1145/3626253.3635573](https://doi.org/10.1145/3626253.3635573).

AI|Sys Amazon SURE Intern

May 2024 – July 2024

Dr. Vishal Misra, Columbia University

New York City, NY

- Developed TokenProbe - an open-source tool for the community to learn how LLMs work on the fly.
- Established workflow diagram on application functionality and design.
- Tested OpenAI's GPT-2 and Meta's LLaMA's 3 4-bit quantized model and tokenizer in Google Cloud VM to investigate in-context learning.
- Demo is hosted by Martin Casado, Marco Mascorro, and Andreessen Horowitz.
- **demo:** <https://tokenprobe.cs.columbia.edu/>.

MM Ronald E. McNair Scholars Researcher

May 2023 – May 2024

Dr. Jeremy Waisome, UF

Gainesville, FL

- Contributed to LithoReality, an Affordable and Realistic Augmented Reality (AR) application.
- Drafted a 3D model to mimic LithoReality's photolithography process.
- Coded a circuitry for the 3D model section for LithoReality's Augmented Reality component.

MM USP Researcher

August 2022 – May 2023

Dr. Jeremiah Blanchard and Audrey DeHoog, UF

Gainesville, FL

- Developed a questionnaire to survey programming fundamental instructors of top 50 largest universities.
- Created a codebook for qualitative coding on survey responses.
- Evaluate instructor responses on programming, collaboration, and classroom style through inter-rater reliability.
- **pub:** <https://peer.asee.org/56440>

MM Engaging Learning Lab Researcher

January 2021 – May 2022

Dr. Cheryl Resch, UF

Gainesville, FL

- Built a codebook based on student responses on reflections before and after assessments.
- Associated themes to each student response based on qualitative coding.
- Conducted thematic analysis and inter-rater reliability with a team of four.
- **pub:** <https://peer.asee.org/40428>

AI UF Data Studio Researcher

May 2023 – May 2024

Dr. Christan Grant & Prof. Alex Webber, UF

Gainesville, FL

- Manually segmented human and creature figures from each Mixtec codex page.
- Documented each cutout based on quality, figure type, and source page.

- Peer reviewed paper for publication submission to AmericasNLP.
- **pub:** americasnlp-1.6.pdf

AI UROP International Intern

May 2022 – July 2022

Dr. Stefan Schiffer, RWTH Aachen University

Aachen, Germany

- Tested facial, mood, eye, upper body and gesture detection OpenCV algorithms.
- Programmed a visual processing algorithm from those previous works to track humans.

AI Leadership Alliance SR-EIP and vASURE Intern

May 2021 – July 2021

Dr. Alberto Quattrini Li, Dartmouth College

Virtual

- Developed a UAV obstacle and landscape detection algorithm to experiment on a ROS simulation.
- Assembled a fundamental obstacle algorithm based on UAV recorded footage.

MM Emerging Scholars Researcher

January 2021 – May 2022

Dr. Jeremiah Blanchard, UF

Gainesville, FL

- Conducted qualitative coding and thematic analysis on students' perspectives of camera-based proctoring systems.
- Worked with inter-rater reliability on the impact of camera-based proctoring systems.

POSTER & ORAL PRESENTATIONS (NOTE: CAN HAVE MULTIPLE PRESENTATIONS)

- **A. Wu**, J. De Chant, T. Hellert, S. Leemann, G. Martino, & A. Sulc. (2026, May). *Evaluating Language Models for Understanding Accelerators at Advanced Light Source*, Poster presented at Research Symposium, Lawrence Berkeley National Laboratory.
- **Wu, A.**, Bissell, A., Li, Y., & Vriza, A. *Magnetic Property Prediction Using Machine Learning in Transition Metal Chalcogenides*. (2025, August). Poster presented at NSF Institute for Data Driven Dynamical Design (ID4) Materials Science Research Symposium, Northwestern University.
(2025, August). Poster presented at "Learning On the Lawn," Argonne National Laboratory.
(2025, July). Oral presentation presented at Department of Energy: Argonne IgniteOff Talks, Argonne National Laboratory.
- **Wu, A.**, Cobb, M., Graddy, N., Perez, V., Grant, C., & Waisome, J. A. M. *Guided Universal Training for Semantic Segmentation (GUTSS)*. (2025, February). Poster presented at Florida Undergraduate Research Conference (FURC), University of South Florida.
(2024, November). Poster presented at the Ronald E. McNair Open House, University of Florida.
(2024, April). Poster presented at the Undergraduate Research Symposium, University of Florida.
(2024, March). Poster presented at ACM SIGCSE 2024. Portland, OR.
(2023, October). Poster presented at the AI Days Symposium, University of Florida.
- Kerfoot, C.*, Rahman, R.*, **Wu, A.***, Tang, A., & Misra, V. (2024, July). *Deeper Dive: How LLMs Learn on the Fly*. Poster presented at the Amazon Summer Undergraduate Research Experience Symposium, Columbia University.
- **Wu, A.**, Parnell Jr., D., & Waisome, J.A.M. *LithoReality: Accessible and Affordable AR Photolithography Application*. (2023, November). Poster presented at the Ronald E. McNair Open House, University of Florida.
(2023, June). Oral presentation presented at Ronald E. McNair SAEOPP Conference, Atlanta, GA.
- **Wu, A.**, DeHoog, A., & Blanchard, J. (2023, April). *Instructor Perspectives on Non-STEM Programming Fundamental Classes*. Poster presented at the Undergraduate Symposium for USP, University of Florida.
- **Wu, A.** & Schiffer, S. (2022, July). *Visual Processing for MoBi, the Monster Bin*. Poster presented at the UROP International Symposium, RWTH Aachen University.
- **Wu, A.**, Stapleton, P., & Blanchard, J. (2022, April). *Student Perspectives on Online Camera-Based Proctoring*. Poster presented at the Undergraduate Symposium for ESP, University of Florida.
- **Wu, A.**, Li, A.Q., & Jeong, M. (2021, July). *UAV Image-Based Obstacle Detection in Aquatic Environments*. Oral presentation presented at the Leadership Alliance National Symposium, Virtual Event.

*All authors contributed equally.

RESEARCH MENTORSHIP

08/2023–05/2025-Natahja Graddy(UF CS BS)(Gathered footage on small animal anatomy dissections for GUTSS)
08/2023–05/2024-Morgan Cobb(UF CS BS)(Drafted application design through Figma & Swift for GUTSS)
08/2023–05/2023-Jinghan (Kevin) Wu(UMich IS BS) (Added features to 3D-model for LithoReality)
06/2021–08/2021-Nathaniel Collins(UF CS BS)(Oversaw peer-reviewed STEPUP work on nanotechnology)
06/2021–08/2021-Keelyn Mooney(UF IE BS) (Oversaw same peer-reviewed STEPUP work)
06/2021–08/2021-Sihala Senevirathne(UF CS BS)(Oversaw same peer-reviewed STEPUP work)
06/2021–08/2021-Katelynn Turney-Rudisill(UF Music Edu BS) (Oversaw same peer-reviewed STEPUP work)

PROFESSIONAL EXPERIENCE FOR TEACHING AND MENTORSHIP

UF: Information and Database Systems 1 Teaching Assistant August 2024 – December 2024
Dr. Alexander Webber, HWCOE, UF *Gainesville, FL*

- Assessed students' level of understanding on their fundamentals in database systems through group projects.
- Developed and implemented a rubric for an assignment's evaluation to maintain consistency within grading.

UF International Center (UFIC): Student Assistant May 2023 – August 2023
Dr. Pingchien Neo, UFIC, UF *Gainesville, FL*

- Oversaw and maintained UFIC site.
- Updated spreadsheet on international programs and class curriculums offered.

WiCSE Shadowing Mentorship Program: Front End Developer Jan. 2023 – April 2023
Traject, WiCSE Shadowing Mentorship Program, UF *Gainesville, FL*

- Designed Figma workflow and React front-end interface for user interaction on a draft for a personal project based on mentorship apps.
- Implemented Firebase back-end for user registration and login.

STEPUP Lead Mentor and Teaching Assistant May 2021 – May 2022
Mr. Stephen Roberts, STEPUP STAR Office, UF *Gainesville, FL*

- Provided academic support to research groups for Dr. Regina Rodriguez for Research Fundamentals.
- Assisted in course management for Dr. Philip Jackson's Engineering Design and Society classes.
- Supervised academic, professional, and personal workshops for first-year engineering students.

STEPUP Participant June 2020 – May 2021
Mr. Stephen Roberts, STEPUP STAR Office, UF *Gainesville, FL*

- Led a mock research project on discrete event simulations for the optimization of patient care during a pandemic.
- Presented a mock oral presentation at the end of the summer program.
- Represented my group in the mock research Q&A session during the research fundamental class.

AWARDS, SCHOLARSHIPS, HONORS, FELLOWSHIPS

10/2024 - **UF: Fernandez Family Scholarship:** Top 10 UF Applicants Interested in Engineering Ph.D.
03/2023 - **McNair SAEOPP Conference 3rd Place Winner:** Chose top 3 oral presenters in education category.
10/2023 - **UF: AI Diversity Fellowship:** 1 of 20 UF high-achieving underrepresented and first-generation low-income UF students and is funded through an NSF grant dedicated to broaden student participation in AI courses and research.
05/2022 - **RWTH Aachen University International UROP Scholarship:** Awarded stipend of 1,850 euros.
05/2022 - **U.S. Department of State: Benjamin A. Gilman Scholarship:** Top 30 % of UF Applicants to study abroad with Pell-grant recipient. Awarded stipend of 5,000 dollars.
04/2022 - **UF: Dr. Ralph Alexander Morgen StepUp Program Endowment:** Awarded 2,000 dollars to high-achieving UF STEPUP participants interested in studying abroad.
08/2021 - **Leadership Alliance: SR-EIP Scholar and Virtual Professional Development Series Certificate:** Awarded to students interested in pursuing a Ph.D. and conducting a research internship at another institution.
09/2020 - **Code for America: Code for Gainesville: National Day of Civic Hacking 1st Place Winner:** Top 3 Project Showcases. Awarded for UI food bank system prototype draft via Figma.
06/2020 - **UF: Machen First-Generation Opportunity Scholarship:** Awarded to high-achieving first-generation, low-income incoming UF students.
06/2020 - **Florida Bright Future Scholarship:** Awarded to high-achieving students in Florida.

OTHER COMPETITIVE ACTIVITIES

05/2026 - Berkeley Lab: Deep Learning for Science Summer School: Top 35% of worldwide applicants.
10/2025 - JHU EHOP: Awarded full funding for recruitment program visit with one-on-one faculty meetups at JHU.
09/2025 - UIUC MERGE: Virtual campus visitation program at UIUC. Also, eligible for \$700 of professional development funds upon UIUC enrollment.
01/2025 - GT FOCUS: Awarded full funding for recruitment program visit at GT.
11/2024 - USC Preview Day Travel Grant: Awarded partial funding for graduate recruitment program at USC.
03/2024 - DAAD RISE: Accepted out of 2,358 applicants; later committed to [Amazon SURE](#).
06/2023 - ConsiderCornell: Explore: Explore graduate school application process.
09/2022 - NSF IOT4AG PPP @ UPenn: Explore graduate school application process and have one-on-one mentorship with UPenn faculty and graduate students.
02/2021 - RWTH Aachen University International UROP: Accepted; later committed to [Leadership Alliance SR-EIP at Dartmouth vASURE](#).

COMMUNITY SERVICE AND OUTREACH

Berkeley Lab Teaching Assistant June 2026 – Present
Intern Volunteer, LBNL *Berkeley, CA*

- Oversaw lectures for highschool students on Python fundamentals and Arduino circuit design.
- Provided feedback on assignments with Python fundamentals and Arduino circuit design.

Berkeley Lab Service Member Jan 2026 – Present
Intern Volunteer, LBNL *Berkeley, CA*

- Spoke about internship experience and research opportunities with community college students (Laney College).
- Discussed platform and dashboard design on open-source user-centered scientific software (BIDS: STRUDEL).

UF: Ronald E. McNair Scholar May 2023 – May 2025
Student Volunteer, UF *Gainesville, FL*

- Shared research expertise and journey at UF BRIDGES Program, an outreach program to prepare Gainesville high school students to be competitive undergraduate applicants.
- Discussed about summer research opportunities on a First Gen Research panel.
- Spoke as a panelist for a class with first-year students on research opportunities at UF.
- Recruited aspiring McNair Scholars at McNair Open House.

UF: International Engineering Ambassadors August 2022 – May 2024
President, HWCOE Student Organization, UF *Gainesville, FL*

- Supervised general body meetings, information sessions, and tabling events for organization.
- Organized trips around Florida (e.g. Kennedy Space Center) for international engineering students.

Department of State: Benjamin A. Gilman Scholar May 2022 – December 2022
Service Project Organizer, HWCOE, UF *Gainesville, FL*

- Led a study abroad panel for UF STEPUP students to learn more about international programs.
- Blogged on study abroad experience.
- Spoke as a study abroad panelist for Germany for UF first-year students.

Miami International Airport (MIA) Airport Ambassador August 2019 – March 2020
Student Volunteer, MIA *Miami, FL*

- Managed the information desk where travelers from over 80 airlines stop by to ask questions.
- Presided over the changes of more than 1000 flights for said travels.
- Oversaw efforts to assist commuters during Hurricane Irma and 3000 passengers for SuperBowl LIV.

TECHNICAL SKILLS

Programming: Java, Python, C/C++, JavaScript
Languages: Cantonese, English, Spanish, German (ask for level of fluency)
Advanced Coursework: Information and Database Systems, AI Fundamentals, Engineering Design and Society